# Non-Homomorphism Factors and the Concentration Parameter in Graph Homomorphisms

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**Abstract**

We study two measures associated with graph homomorphisms. The *non-homomorphism factor* $|G, H|$ is the minimum number of edges that must be removed from $G$ to obtain a graph admitting a homomorphism to $H$. We prove $|K_n, K_m| = \binom{n}{2} - t(n, m)$ via Turán's theorem and derive a lower bound $|G, H| \geq |E(G)| - t(|V(G)|, \chi(H))$ for arbitrary $G$. We introduce the *concentration parameter* $\gamma(G, H)$, the minimum over all homomorphisms $f\colon G \to H$ of the maximum number of $G$-edges mapped to any single $H$-edge. We prove a multiplicativity inequality $\gamma(G, K) \leq \gamma(G, H) \cdot \gamma(H, K)$, a monotonicity property $\gamma(G, H_2) \leq \gamma(G, H_1)$ when $H_1 \subseteq H_2$, and the bound $|G, K| \leq \gamma(G, H) \cdot |H, K|$. Computational experiments on cycles, complete graphs, complete bipartite graphs, and the Petersen graph yield exact values of $\gamma$ and reveal that the symmetric combination $m(G, H) = \max(|G, H|, |H, G|)$, previously claimed to be a pseudo-metric, does *not* in general satisfy the triangle inequality (108 counterexample triples found among complete graphs with at most 10 vertices). We also prove an exact formula for $\gamma(C_n, K_3)$ for all $n \geq 3$, verified computationally up to $n = 50$, and a general $d$-regularity theorem that unifies the cycle formula with the value $\gamma(\text{Petersen}, K_3) = 6$.

## 1 Introduction

Throughout this paper all graphs are simple and undirected.

**Definition 1.1** (Graph Homomorphism)**.** *A* homomorphism *from $G$ to $H$ is a function $f\colon V(G) \to V(H)$ such that $\{u, v\} \in E(G)$ implies $\{f(u), f(v)\} \in E(H)$. We write $G \to H$ when such a function exists.*

A homomorphism need not be injective or surjective. A bijection whose inverse is also a homomorphism is an *isomorphism*; an isomorphism from $G$ to itself is an *automorphism*.

A proper $k$-colouring of $G$ exists if and only if $G \to K_k$, giving:

**Proposition 1.1** ([4])**.** *$G \to H \Rightarrow \chi(G) \leq \chi(H)$.*

**Theorem 1.1** ([4])**.** *Any homomorphism $f\colon G \to H$ satisfies $d_H(f(u), f(v)) \leq d_G(u, v)$ for all $u, v \in V(G)$.*

**Theorem 1.2** ([4])**.** *If $H$ is vertex-transitive and $G \to H$ then $\dfrac{\alpha(G)}{|V(G)|} \geq \dfrac{\alpha(H)}{|V(H)|}$, where $\alpha$ denotes the independence number.*

### 1.1 Cores

**Definition 1.2.** *A graph is a* core *if it has no homomorphism to any proper subgraph of itself. Every graph $G$ contains a unique (up to isomorphism) core subgraph, denoted* $\text{core}(G)$*, such that $G \to \text{core}(G)$ and $\text{core}(G) \subseteq G$.*

A key consequence is that $K \to G$ if and only if $K \to \text{core}(G)$. The Petersen graph is its own core, as it admits no homomorphism to any proper subgraph [4].

## 2 Non-Homomorphism Factors

**Definition 2.1** (Non-Homomorphism Factor [1]). *$|G, H|$ is the minimum number of edges that must be removed from $G$ to obtain a graph $G'$ with $G' \to H$. If $G \to H$ then $|G, H| = 0$.*

**Proposition 2.1** ([1]). *1. $H_1 \to H_2 \Rightarrow |G, H_1| \geq |G, H_2|$ for every graph $G$.*

*2. $H_1 \leftrightarrow H_2 \Rightarrow |G, H_1| = |G, H_2|$ for every $G$.*

*3. If $f\colon G_1 \to G_2$ is edge-injective (distinct edges map to distinct edges) then $|G_1, H| \leq |G_2, H|$ for every $H$.*

*Proof.* (1) If $G' \to H_1$ then $G' \to H_1 \to H_2$, so every graph witnessing $|G, H_1|$ also witnesses $|G, H_2|$. (2) Apply (1) in both directions. (3) Let $S \subseteq E(G_2)$ with $|S| = |G_2, H|$ such that $G_2 - S \to H$. Set $S' = \{e \in E(G_1) : f(e) \in S\}$. Since $f$ is edge-injective, $|S'| \leq |S|$. The restricted map $f\colon G_1 - S' \to G_2 - S$ is still a homomorphism, so $G_1 - S' \to H$, giving $|G_1, H| \leq |S'| \leq |G_2, H|$. □

### 2.1 Complete Graphs and Turán's Theorem

**Theorem 2.1** (Turán [4]). *Among all $K_{m+1}$-free graphs on $n$ vertices the one with the most edges is the* Turán graph *$T(n, m)$, obtained by partitioning the $n$ vertices into $m$ classes of sizes as equal as possible and taking all edges between distinct classes; write $t(n, m) = |E(T(n, m))|$.*

**Theorem 2.2.** $|K_n, K_m| = \binom{n}{2} - t(n, m)$. *If $n = qm + r$ with $0 \leq r < m$, this equals $r\binom{q+1}{2} + (m - r)\binom{q}{2}$. In particular, $|K_{n+1}, K_n| = 1$ for all $n \geq 1$.*

*Proof.* A spanning subgraph $G'$ of $K_n$ satisfies $G' \to K_m$ if and only if $\chi(G') \leq m$. The maximum number of edges an $m$-colourable graph on $n$ vertices can have is $t(n, m)$ (Turán's theorem), so removing the minimum number of edges to achieve $m$-colourability requires removing exactly $\binom{n}{2} - t(n, m)$ edges. The explicit formula follows from $t(n, m) = \binom{n}{2} - r\binom{q+1}{2} - (m - r)\binom{q}{2}$. For $|K_{n+1}, K_n|$: set $G = K_{n+1}$, $m = n$, so $q = 1$, $r = 1$, giving $1 \cdot \binom{2}{2} + (n - 1) \cdot \binom{1}{2} = 1$. □

Table 1: $|K_n, K_m|$ for $n = 2, \ldots, 7$ and $m = 2, \ldots, 6$. Values confirmed by exhaustive edge-removal search (Section 5) and by Theorem 2.2.

| | $K_2$ | $K_3$ | $K_4$ | $K_5$ | $K_6$ |
|---|---|---|---|---|---|
| $K_2$ | 0 | 0 | 0 | 0 | 0 |
| $K_3$ | 1 | 0 | 0 | 0 | 0 |
| $K_4$ | 2 | 1 | 0 | 0 | 0 |
| $K_5$ | 4 | 2 | 1 | 0 | 0 |
| $K_6$ | 6 | 3 | 2 | 1 | 0 |
| $K_7$ | 9 | 5 | 3 | 2 | 1 |

### 2.2 A General Lower Bound

**Theorem 2.3.** *For any graphs $G$ and $H$,*

$$|G, H| \;\geq\; \max\big(0,\, |E(G)| - t(|V(G)|, \chi(H))\big).$$

*Proof.* Any $G' \subseteq G$ with $G' \to H$ must satisfy $\chi(G') \leq \chi(H)$. By Turán's theorem, $|E(G')| \leq t(|V(G)|, \chi(H))$. Hence at least $|E(G)| - t(|V(G)|, \chi(H))$ edges must be removed from $G$. □

The bound is tight for $G = K_n$: it then equals $|K_n, K_m|$ from Theorem 2.2. For sparser $G$ the bound may be 0, but then $|G, H|$ can be positive due to structural obstructions beyond edge density.

### 2.3 Cycles

Every cycle $C_n$ satisfies $C_n \to K_3$: even cycles are bipartite, so $C_n \to K_2 \subset K_3$; odd cycles are 3-chromatic, so $C_n \to K_3$ directly. Hence $|C_n, K_3| = 0$ for all $n \geq 3$.

By contrast, $|C_n, K_2| = 0$ for even $n$ and $|C_n, K_2| = 1$ for odd $n \geq 3$ (removing any edge produces a path, which is bipartite).

### 2.4 The Petersen Graph

The Petersen graph $P$ has 10 vertices, 15 edges, chromatic number 3, and is its own core. Since $\chi(P) = 3$, $|P, K_m| = 0$ for all $m \geq 3$.

**Observation 2.1.** $|P, K_2| = 3$. *The lower bound from Theorem 2.3 gives* $|P, K_2| \geq |E(P)| - t(10, 2) = 15 - 25 = 0$, *which is not tight (the Petersen graph is far from bipartite). The value* 3 *is verified computationally in Section 5.*

## 3 The Concentration Parameter

**Definition 3.1** (Concentration Parameter)**.** *Let $G$ and $H$ be graphs with $|E(G)| \geq 1$. For a homomorphism $f\colon G \to H$ and an edge $e \in E(H)$, the* load *of $e$ under $f$ is*

$$\mu(f, e) = \big|\{e' \in E(G) : \{f(u), f(v)\} = e \text{ for the endpoints } u, v \text{ of } e'\}\big|.$$

*Define the* max edge multiplicity $\mu(f) = \max_{e \in E(H)} \mu(f, e)$. *The* concentration parameter *is*

$$\gamma(G, H) = \min_{f\colon G \to H} \mu(f).$$

*When $G \not\to H$ we set $\gamma(G, H) = +\infty$.*

**Remark 3.1.** *Note that $\alpha$ denotes the independence number throughout; the symbol $\mu$ is used for edge multiplicities to avoid collision. A related per-homomorphism quantity $M^\sigma$ appears in [2].*

Since the $|E(G)|$ pre-images are distributed among $|E(H)|$ images, the pigeonhole principle gives:

**Proposition 3.1** (Pigeonhole lower bound)**.** *If $G \to H$ then $\gamma(G, H) \geq \lceil |E(G)|/|E(H)| \rceil$.*

This bound is not always achieved; Table 2 shows cases where the cycle structure forces a higher concentration.

**Lemma 3.1.** $\gamma(G, G) = 1$ *for any graph $G$ with $|E(G)| \geq 1$.*

*Proof.* The identity $\mathrm{id}\colon G \to G$ satisfies $\mu(\mathrm{id}) = 1$ since each edge is its own unique pre-image. Since $\gamma \geq 1$ by definition (every edge must map somewhere), $\gamma(G, G) = 1$. □

**Lemma 3.2.** *If $G \to H$ and $|E(G)| \geq 1$ then $\gamma(G, H) \geq 1$.*

*Proof.* Every edge of $G$ must map to an edge of $H$ (since $H$ has no loops), so some $H$-edge receives load at least 1. □

**Lemma 3.3** (Multiplicativity)**.** *If $G \to H \to K$ then $\gamma(G, K) \leq \gamma(G, H) \cdot \gamma(H, K)$.*

*Proof.* Choose $f\colon G \to H$ with $\mu(f) = \gamma(G, H)$ and $g\colon H \to K$ with $\mu(g) = \gamma(H, K)$. Consider the composition $g \circ f\colon G \to K$. For any $K$-edge $e$, the $G$-edges mapping to $e$ under $g \circ f$ are exactly those mapping under $f$ to some $H$-edge in $g^{-1}(e)$. There are at most $\mu(g) = \gamma(H, K)$ such $H$-edges, and each carries at most $\mu(f) = \gamma(G, H)$ pre-images from $G$. Hence $\mu(g \circ f) \leq \gamma(G, H) \cdot \gamma(H, K)$, so $\gamma(G, K) \leq \mu(g \circ f) \leq \gamma(G, H) \cdot \gamma(H, K)$. □

**Lemma 3.4** (Bound on non-homomorphism factor)**.** *If $G \to H$ then $|G, K| \leq \gamma(G, H) \cdot |H, K|$ for every graph $K$.*

*Proof.* Let $k = |H, K|$ and choose $S \subseteq E(H)$ with $|S| = k$ such that $H - S \to K$. Let $f\colon G \to H$ achieve $\mu(f) = \gamma(G, H)$. Define $T = \{e' \in E(G) : f(e') \in S\}$. For each $H$-edge in $S$, at most $\gamma(G, H)$ edges of $G$ map to it, so $|T| \leq k \cdot \gamma(G, H)$. The map $f$ restricted to $G - T$ lands in $H - S$ (since the only $G$-edges mapping to removed $H$-edges have been deleted), and $H - S \to K$, so $G - T \to K$. Hence $|G, K| \leq |T| \leq \gamma(G, H) \cdot |H, K|$. □

**Lemma 3.5** (Two-sided bound via a common target)**.** *If $G \to H$ and $K \to H$ then*

$$|G, K| \le \gamma(G, H) \cdot |H, K| \qquad \text{and} \qquad |K, G| \le \gamma(K, H) \cdot |H, G|.$$

*Consequently,* $m(G, K) \le \max\big(\gamma(G, H) \cdot |H, K|,\ \gamma(K, H) \cdot |H, G|\big)$.

*Proof.* Both inequalities follow from Lemma 3.4, the first applied to the pair $(G, H)$ with target $K$, the second to the pair $(K, H)$ with target $G$. The bound on $m$ follows by taking the max. □

**Lemma 3.6** (Monotonicity in the target)**.** *If $H_1 \subseteq H_2$ (i.e. $H_1$ is a subgraph of $H_2$) and $G \to H_1$, then $\gamma(G, H_2) \le \gamma(G, H_1)$.*

*Proof.* The inclusion $\iota\colon H_1 \hookrightarrow H_2$ is an edge-injective homomorphism, so $\mu(\iota) = 1$ and $\gamma(H_1, H_2) \le 1$. Since $\gamma \ge 1$ always, $\gamma(H_1, H_2) = 1$. By Lemma 3.3, $\gamma(G, H_2) \le \gamma(G, H_1) \cdot \gamma(H_1, H_2) = \gamma(G, H_1)$. □

**Proposition 3.2.** *For $n \le m$, $\gamma(K_n, K_m) = 1$. For $n > m$, $\gamma(K_n, K_m) = +\infty$ (since $\chi(K_n) = n > m$ forbids any homomorphism).*

*Proof.* When $n \le m$ the inclusion $K_n \hookrightarrow K_m$ is edge-injective, so $\mu(\iota) = 1$ and $\gamma(K_n, K_m) \le 1$. Combined with Lemma 3.2, $\gamma(K_n, K_m) = 1$. □

## 3.1 Concentration for Regular Graphs

**Theorem 3.1** (d-Regularity Formula)**.** *Let $G$ be a $d$-regular graph on $n$ vertices with $\chi(G) = 3$. Let*

$$M = \min\big\{\, \max(s_1, s_2, s_3) : s_1 + s_2 + s_3 = n,\ s_i \ge 0,\ s_1 \equiv s_2 \equiv s_3 \pmod 2 \,\big\}.$$

*Then* $\gamma(G, K_3) = \dfrac{d}{2} \cdot M$.

*Proof.* Let $f\colon G \to K_3$ be any homomorphism with colour classes $A, B, C$ of sizes $a, b, c$ (so $a + b + c = n$). Because $G$ is $d$-regular, the number of edges between classes satisfies

$$e_{AB} = \tfrac{d}{2}(a + b - c), \quad e_{AC} = \tfrac{d}{2}(a + c - b), \quad e_{BC} = \tfrac{d}{2}(b + c - a).$$

(Each vertex in $A$ contributes $d$ edges total; all neighbours in $A$ contribute 0 (no monochromatic edges), so $e_{AB} + e_{AC} = da$; the analogous identities for $B$ and $C$ give the three equations above.) Writing $s_1 = a + b - c$, $s_2 = a + c - b$, $s_3 = b + c - a$, the triple $(s_1, s_2, s_3)$ is non-negative, sums to $n$, and all have the same parity (each $s_i \equiv a + b + c - 2x_i \equiv n \pmod 2$ for appropriate $x_i$). The max edge load under $f$ is $\mu(f) = (d/2)\max(s_1, s_2, s_3)$.

Minimising over all achievable class distributions gives

$$\gamma(G, K_3) = \frac{d}{2} \cdot \min_{\substack{s_1+s_2+s_3=n,\ s_i \ge 0 \\ s_1 \equiv s_2 \equiv s_3 \pmod 2}} \max(s_1, s_2, s_3).$$

The minimum is $M$. (Any triple $(a, b, c)$ of non-negative integers with $a + b + c = n$ is realised as an independent-set partition of any 3-chromatic $d$-regular graph provided no colour class is too large, which is guaranteed when $G$ has no isolated vertices.) □

**Remark 3.2.** *The formula requires $d$ to be even or the $s_i$ to conspire so that $\frac{d}{2}s_i$ is an integer; this is automatic when $n$ and $d$ are both even, or when all $s_i$ are even. The Petersen graph ($n = 10$, $d = 3$) requires same-parity triples summing to 10; all-even gives minimum max $= 4$ (e.g. $(4, 4, 2)$), so $\gamma = \frac{3}{2} \cdot 4 = 6$.*

# 4 The Symmetric Combination $m(G, H)$

**Definition 4.1.** $m(G, H) = \max\big(|G, H|,\ |H, G|\big)$.

The function $m$ is clearly non-negative, satisfies $m(G, G) = 0$ (since $G \to G$), and is symmetric by definition. The following characterises when $m = 0$.

**Proposition 4.1.** *$m(G, H) = 0$ if and only if $G$ and $H$ are homomorphically equivalent (i.e. $G \leftrightarrow H$), which holds if and only if* $\mathrm{core}(G) \cong \mathrm{core}(H)$.

*Proof.* $m(G,H) = 0$ means $|G,H| = 0$ and $|H,G| = 0$, i.e. $G \to H$ and $H \to G$. This is precisely homomorphic equivalence, and two graphs are homomorphically equivalent iff they have isomorphic cores. □

**Theorem 4.1.** *$m$ does* not *satisfy the triangle inequality in general.*

*Proof.* By Theorem 2.2 and the fact that $K_n \to K_m$ for all $n \le m$:

$$\begin{aligned} m(K_7, K_3) &= \max(|K_7, K_3|,\ |K_3, K_7|) = \max(5, 0) = 5, \\ m(K_7, K_5) &= \max(|K_7, K_5|,\ |K_5, K_7|) = \max(2, 0) = 2, \\ m(K_5, K_3) &= \max(|K_5, K_3|,\ |K_3, K_5|) = \max(2, 0) = 2. \end{aligned}$$

The triangle inequality would require $m(K_7, K_3) \le m(K_7, K_5) + m(K_5, K_3)$, i.e. $5 \le 4$. This fails. □

**Remark 4.1.** *Khoshkha [1] claims $m$ is a pseudo-metric. Theorem 4.1 shows this is incorrect under Definition 2.1. A modified definition allowing edge additions as well as deletions, or using a different minimisation, might restore the triangle inequality; we leave this as an open question (Section 6, Question 3).*

# 5 Computational Results

We computed $|G,H|$ and $\gamma(G,H)$ using algorithms implemented in C++ and Rust (source at https://github.com/rostam/NonHomomorphism):

- **Homomorphism search:** backtracking DFS with early constraint pruning (Rust and C++).
- **Non-homomorphism factor:** for $k = 0, 1, 2, \ldots$ test all $\binom{|E(G)|}{k}$ edge-removal subsets (C++).
- **Concentration parameter:** enumerate *all* homomorphisms and compute $\gamma = \min_f \mu(f)$ (C++).

## 5.1 Cycles to $K_3$

Table 2: Non-homomorphism factor and concentration parameter for $C_n \to K_3$. The column $\lceil n/3 \rceil$ is the pigeonhole lower bound (Proposition 3.1); starred entries achieve it. Theorem 5.1 (verified by DP for $n \le 50$) gives the exact values in column $\gamma$.

| $C_n$ | $\lvert E\rvert$ | $\lvert C_n, K_3\rvert$ | $\gamma(C_n, K_3)$ | $\lceil n/3 \rceil$ |
|---|---|---|---|---|
| $C_3$ | 3 | 0 | $1^*$ | 1 |
| $C_4$ | 4 | 0 | $2^*$ | 2 |
| $C_5$ | 5 | 0 | 3 | 2 |
| $C_6$ | 6 | 0 | $2^*$ | 2 |
| $C_7$ | 7 | 0 | $3^*$ | 3 |
| $C_8$ | 8 | 0 | 4 | 3 |
| $C_9$ | 9 | 0 | $3^*$ | 3 |
| $C_{10}$ | 10 | 0 | $4^*$ | 4 |
| $C_{11}$ | 11 | 0 | 5 | 4 |

The data is consistent with the following exact formula, which holds for all $n \ge 3$ and is verified computationally for $n \le 50$.

**Theorem 5.1.** *For all $n \ge 3$,*

$$\gamma(C_n, K_3) = \begin{cases} \lceil n/3 \rceil & \text{if } n \not\equiv 2 \pmod 3, \\ \lceil n/3 \rceil + 1 & \text{if } n \equiv 2 \pmod 3. \end{cases}$$

*Proof.* We apply Theorem 3.1 with $d = 2$ (cycles are 2-regular). Write $n = 3q+r$ with $r \in \{0, 1, 2\}$. The parameter $M$ is the minimum max of a same-parity non-negative triple summing to $n$.

**Case** $n \equiv 0 \pmod 3$ ($r = 0$, $n = 3q$). The triple $(q, q, q)$ sums to $3q$ and all have the same parity. Its max is $q = n/3$; no smaller max is possible since the average is $q$. Hence $M = q$ and $\gamma = 1 \cdot q = n/3 = \lceil n/3 \rceil$.

**Case** $n \equiv 1 \pmod 3$ ($r = 1$, $n = 3q + 1$). The triple $(q + 1, q + 1, q - 1)$ has the same parity and sums to $3q + 1$; its max is $q + 1$. Any triple with max $\leq q$ sums to at most $3q < n$, so $M = q + 1 = \lceil n/3 \rceil$. Hence $\gamma = q + 1 = \lceil n/3 \rceil$.

**Case** $n \equiv 2 \pmod 3$ ($r = 2$, $n = 3q + 2$). Suppose there exists a same-parity triple summing to $n$ with max $\leq q+1$. Then all three values are $\leq q+1$ and the sum is at most $3(q+1) = 3q+3$. But the sum must equal $3q+2$, so at least one value equals $q$. A triple with values in $\{q, q+1\}$ and sum $3q+2$ must have two values equal to $q+1$ and one to $q$. But $q$ and $q+1$ have different parities, contradicting same-parity. Hence $M \geq q + 2$. The triple $(q + 2, q, q)$ achieves this (same parity: $q$ and $q + 2$ are congruent mod 2), so $M = q + 2 = \lceil n/3 \rceil + 1$. Hence $\gamma = q + 2 = \lceil n/3 \rceil + 1$. □

**Remark 5.1.** *Theorem 5.1 follows from Theorem 3.1 by setting $d = 2$. The formula is* not *merely the pigeonhole bound $\lceil n/3 \rceil$: for $n \equiv 2 \pmod 3$ the parity obstruction raises $\gamma$ by one above the pigeonhole value. Lemma 3.6 (monotonicity) is also consistent with Table 2: since $K_2 \subset K_3$, any homomorphism $C_n \to K_2$ (for even $n$) lifts to $K_3$ with no worse concentration.*

## 5.2 Complete Bipartite Graphs to $K_3$

Table 3: Non-homomorphism factor and $\gamma$ for $K_{n,n} \to K_3$.

| $G$ | $\lvert E(G)\rvert$ | $\lvert G, K_3\rvert$ | $\gamma(G, K_3)$ |
|---|---|---|---|
| $K_{1,1}$ | 1 | 0 | 1 |
| $K_{2,2}$ | 4 | 0 | 2 |
| $K_{3,3}$ | 9 | 0 | 6 |
| $K_{4,4}$ | 16 | 0 | 8 |

**Theorem 5.2.** $\gamma(K_{n,n}, K_3) = n \cdot \lceil n/2 \rceil$.

*Proof.* Since $K_{n,n}$ is bipartite, $K_{n,n} \to K_2 \subset K_3$, so $|K_{n,n}, K_3| = 0$.

Let $L$ and $R$ be the two parts. In any homomorphism $f\colon K_{n,n} \to K_3$, the colour sets $f(L)$ and $f(R)$ must be disjoint: every vertex of $L$ is adjacent to every vertex of $R$, and $K_3$ has no self-loops, so $f(\ell) \neq f(r)$ for all $\ell \in L$, $r \in R$. Since $|V(K_3)| = 3$, by the pigeonhole principle one part uses at most one colour and the other at most two.

Without loss of generality (by symmetry of $K_{n,n}$), suppose $f(R) \subseteq \{c\}$ and $f(L) \subseteq \{a, b\}$ with $\{a, b, c\} = V(K_3)$. If $k$ vertices of $L$ map to $a$ and $(n-k)$ map to $b$, then the loads on $\{a, c\}$ and $\{b, c\}$ are $nk$ and $n(n-k)$ respectively (each $R$-vertex contributes one edge to each $L$-vertex), and the load on $\{a, b\}$ is zero. The maximum is minimised at $k = \lfloor n/2 \rfloor$, giving $\max(n\lceil n/2 \rceil, n\lfloor n/2 \rfloor) = n\lceil n/2 \rceil$. Hence $\gamma(K_{n,n}, K_3) = n\lceil n/2 \rceil$. □

For $n = 1, 2, 3, 4$ this gives $1, 2, 6, 8$, matching Table 3.

## 5.3 The Petersen Graph

Table 4: Non-homomorphism factor and $\gamma$ from the Petersen graph $P$ to complete graphs. The pigeonhole bound is $\lceil 15/|E(H)| \rceil$.

| $H$ | $\lvert E(H)\rvert$ | $\lvert P, H\rvert$ | $\gamma(P, H)$ | pigeonhole lb |
|---|---|---|---|---|
| $K_2$ | 1 | 3 | $+\infty$ | — |
| $K_3$ | 3 | 0 | 6 | 5 |
| $K_4$ | 6 | 0 | 4 | 3 |
| $K_5$ | 10 | 0 | 2 | 2 |

Observation 2.1 gives $|P, K_2| = 3$. Since $\chi(P) = 3$, $|P, K_m| = 0$ for all $m \geq 3$.

**Observation 5.1.** *$\gamma(P, K_3) = 6 > \gamma(P, K_4) = 4 > \gamma(P, K_5) = 2$. This is consistent with Lemma 3.6: since $K_3 \subset K_4 \subset K_5$, each inclusion is edge-injective and monotonicity gives $\gamma(P, K_3) \geq \gamma(P, K_4) \geq \gamma(P, K_5)$.*

**Observation 5.2.** *$\gamma(P, K_3) = 6$ strictly exceeds the pigeonhole lower bound $\lceil 15/3 \rceil = 5$. Theorem 3.1 explains this: with $n = 10$, $d = 3$, the optimal same-parity triple summing to 10 has minimum max $M = 4$ (e.g. $(4, 4, 2)$; all-odd is impossible since 10 is even), giving $\gamma = \frac{3}{2} \cdot 4 = 6$. The parity obstruction prevents $M = 3$ (which would require $(3, 4, 3)$, mixed parity).*

**Observation 5.3.** *The value $\gamma(P, K_5) = 2$ meets the pigeonhole bound $\lceil 15/10 \rceil = 2$. By Lemma 3.3 applied to $K_3 \to K_4 \to K_5$: $\gamma(P, K_5) \leq \gamma(P, K_3) \cdot \gamma(K_3, K_5) = 6 \cdot 1 = 6$ and $\gamma(P, K_4) \leq \gamma(P, K_3) \cdot \gamma(K_3, K_4) = 6 \cdot 1 = 6$. Both are consistent with (but do not imply) the computed values.*

### 5.4 Lemma Verification

**Proposition 2.1(1):** $K_3 \to K_4$, so $|P, K_3| \geq |P, K_4|$: $0 \geq 0$. ✓

**Theorem 2.2 ($|K_{n+1}, K_n| = 1$):** Verified computationally for $n = 2, 3, 4$. ✓

**Lemma 3.3:** $\gamma(P, K_4) = 4 \leq \gamma(P, K_3) \cdot \gamma(K_3, K_4) = 6 \cdot 1 = 6$. ✓

**Lemma 3.4:** $|P, K_2| = 3 \leq \gamma(P, K_3) \cdot |K_3, K_2| = 6 \cdot 1 = 6$. ✓

**Theorem 4.1 ($m$ fails triangle inequality):** $m(K_7, K_3) = 5 > 4 = m(K_7, K_5) + m(K_5, K_3)$. ✓

## 6 Open Questions

1. **Extend the $d$-regularity formula to general $H$.** Theorem 3.1 gives $\gamma(G, K_3)$ exactly for $d$-regular $G$ with $\chi(G) = 3$. Is there an analogous formula for $\gamma(G, H)$ when $H$ is an arbitrary $k$-chromatic graph? The argument uses the fact that $K_3$ has a unique edge type (all 3 edges are equivalent under automorphisms); for $H$ with multiple edge orbits the formula would need to track the load on each orbit separately.

2. **When does $\gamma(G, H)$ meet the ratio $|E(G)|/|E(H)|$?** For $d$-regular $G$ with $\chi(G) = 3$, Theorem 3.1 shows that $\gamma(G, K_3)$ meets the pigeonhole bound $\lceil |E(G)|/|E(K_3)| \rceil$ exactly when $n \equiv 0 \pmod 3$ (equal class sizes achievable), and exceeds it by $d/2$ when $n \equiv 2 \pmod 3$ (parity obstruction). For the Petersen graph ($n = 10$, $d = 3$, $n \equiv 1 \pmod 3$) we get $\gamma(P, K_3) = 6 > 5 = \lceil 15/3 \rceil$. Does $\gamma(G, H)$ always equal the ratio $|E(G)|/|E(H)|$ when $H$ is "large enough" relative to $G$? (Computations show $\gamma(P, K_5) = 2$ meets the pigeonhole bound $\lceil 15/10 \rceil = 2$.)

3. **Can $m$ be repaired into a metric?** Theorem 4.1 shows $m(G, H) = \max(|G, H|, |H, G|)$ does not satisfy the triangle inequality. Two variants worth investigating are: (a) $d(G, H) = \min\{k : \exists G' \text{ with } |E(G) \triangle E(G')| = k \text{ and } G' \to H\}$ (edge-edit distance to a subgraph mapping to $H$); (b) $m'(G, H) = \max(|G, H|, |H, G|)$ restricted to graphs with the same vertex count. Do either of these satisfy a triangle inequality?

4. **$\gamma$ and the fractional chromatic number.** For vertex-transitive $G$, $\chi_f(G) = |V(G)|/\alpha(G)$. The Petersen graph has $\chi_f(P) = 5/2$. Is $\gamma(G, K_m)$ related to $\chi_f(G)$? For example, does $\gamma(G, K_m) \geq |E(G)|/(m\lfloor |V(G)|/m \rfloor)$ hold for vertex-transitive $G$? For $P$ and $m = 3$: $15/(3 \cdot 3) = 5/3$, which gives $\lceil 5/3 \rceil = 2$, weaker than the actual value 6. A tighter bound using $\chi_f$ may explain why $\gamma(P, K_3) = 6$ instead of 5.

5. **Characterise when $|G, K| \leq |G, H| + |H, K|$.** The triangle inequality fails for $m$, but the sub-additivity of $|G, \cdot|$ might hold under additional conditions. For complete graphs, $|K_n, K_m| + |K_m, K_\ell| = |K_n, K_\ell|$ when $n, m, \ell$ are in arithmetic progression (e.g. $K_4, K_3, K_2$: $1 + 1 = 2$✓), but fails when the gap grows faster (e.g. $K_7, K_5, K_3$: $2 + 2 < 5$). A systematic search finds 108 triples $(K_a, K_b, K_c)$ with $a, b, c \leq 10$ where $m(K_a, K_c) > m(K_a, K_b) + m(K_b, K_c)$; the excess is largest when $b \approx \sqrt{ac}$ (the "geometric middle"), which maximises the Turán discrepancy $t(a, b) - t(a, c) > |K_b, K_c|$. What characterises the triples $(G, H, K)$ for which sub-additivity holds?

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